

FICTIONAL DEMO SUBMISSION

For LMS grading-workflow demonstration only

Assignment 2 - Polynomial and Rational Functions

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Selected responses

These original responses are intentionally concise and are provided only to demonstrate Moodle submission, download, and grading workflows.

Question 2

For $F(x) = 2x^3 - 5x^2 - 4x + 3$, $F(3) = 54 - 45 - 12 + 3 = 0$. By the factor theorem, $x - 3$ is a factor.

Question 3

Grouping gives $x^2(x + 2) - 9(x + 2) = (x + 2)(x - 3)(x + 3)$. The intercepts are $x = -3, -2, 3$ and $y = -18$. Since the leading coefficient is positive and the degree is odd, the graph falls left and rises right.

Question 6

The dimensions require $x > 0$, $30 - 2x > 0$, and $22 - 2x > 0$, so $0 < x < 11$. The model ignores material thickness and assumes perfectly square cuts and folds.
